

```
127 • select * from emp_record_table; --
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128
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130
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Result Grid Filter Rows: Export: Wrap Cell Content:											
	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_R
▶	E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5
	E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3
	E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2
	E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5
	E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1
	E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5
	E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4
	E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5
	E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2

```
154 • select * from data_science_team ;
```

Result Grid Filter Rows: Export: Wrap Cell Content:									
	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT
▶	E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA
	E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE
	E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA
	E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA
	E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE
	E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA
	E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA
	E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA
	E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA

data_science_team 8 x Read Only

```
140 • describe Proj_table;
```

```
141 • select * from Proj_table ;
```

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142
```

Result Grid Filter Rows: Export: Wrap Cell Content:							
	PROJECT_ID	PROJ_Name	DOMAIN	START_DATE	CLOSURE_DATE	DEV_QTR	STATUS
▶	P103	Drug Discovery	HEALTHCARE	2021-06-04	2021-06-06	Q1	DONE
	P105	Fraud Detection	FINANCE	2021-11-04	2021-11-06	Q1	DONE
	P109	Market Basket Analysis	RETAIL	2021-12-04	2021-12-06	Q1	DELAYED
	P204	Supply Chain Management	AUTOMOTIVE	2021-12-07	2021-12-09	Q2	WIP
	P302	Early Detection of Lung Cancer	HEALTHCARE	2021-08-10	2021-08-12	Q3	YTS
	P406	Customer Sentiment Analysis	RETAIL	2021-09-07	2021-09-09	Q2	WIP

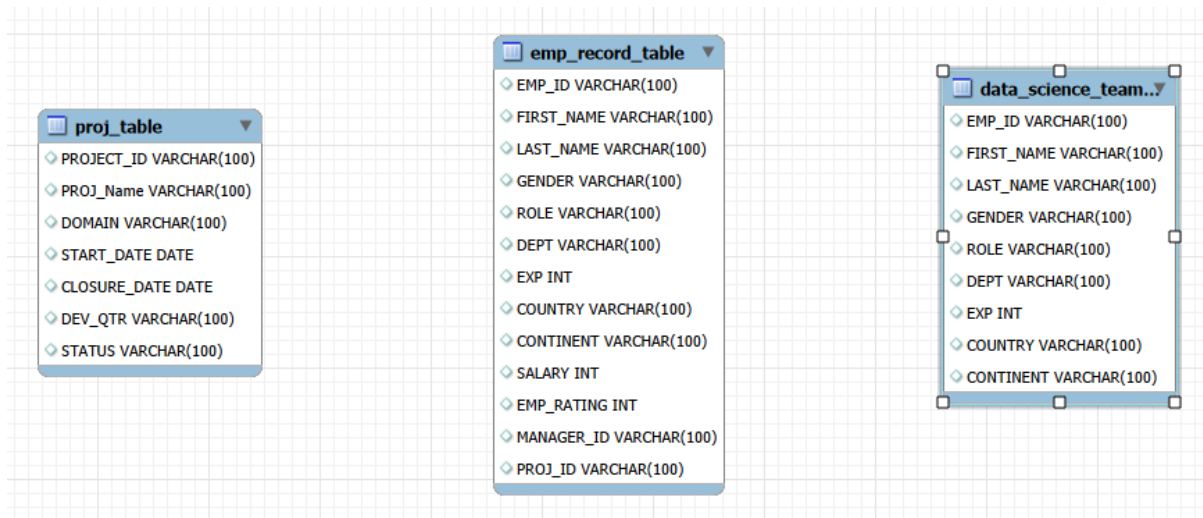
Proj_table 7 x Read Only

1.

Create a database named employee, then import data_science_team.csv proj_table.csv and emp_record_table.csv into the employee database from the given resources.

2.

Create an ER diagram for the given employee database.



3. Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

```

18 • select EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT
19   from emp_record_table;
20
21 • SELECT EMP_ID, FIRST_NAME, LAST_NAME , DEPT
22   FROM emp_record_table;
23

```

Result Grid

	EMP_ID	FIRST_NAME	LAST_NAME	DEPT
▶	E001	Arthur	Black	ALL
	E005	Eric	Hoffman	FINANCE
	E010	William	Butler	AUTOMOTIVE
	E052	Dianna	Wilson	HEALTHCARE
	E057	Dorothy	Wilson	HEALTHCARE

4. Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPARTMENT, and EMP_RATING if the EMP_RATING is: ● ● ● 5. less than two greater than four between two and four.

```

24 • SELECT
25     EMP_ID,
26     FIRST_NAME,
27     LAST_NAME,
28     GENDER,
29     DEPT,
30     EMP_RATING
31 FROM emp_record_table
32 WHERE EMP_RATING < 2
33     OR EMP_RATING > 4
34     OR EMP_RATING BETWEEN 2 AND 4;
35

```

Result Grid Filter Rows: Export: Wrap Cell Content:						
	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT	EMP_RATING
▶	E001	Arthur	Black	M	ALL	5
	E005	Eric	Hoffman	M	FINANCE	3
	E010	William	Butler	M	AUTOMOTIVE	2
	E052	Dianna	Wilson	F	HEALTHCARE	5
	E057	Dorothy	Wilson	F	HEALTHCARE	1
	E082	Patrick	Veltz	M	HEALTHCARE	5

5. Write a query to concatenate the FIRST_NAME and the LAST_NAME of employees in the Finance department from the employee table and then give the resultant column alias as NAME.

```

36 • SELECT
37     CONCAT(FIRST_NAME, ' ', LAST_NAME) AS NAME
38 FROM emp_record_table
39 WHERE DEPT = 'Finance';

```

Result Grid Filter Rows: Export: Wrap Cell Content:	
	NAME
▶	Eric Hoffman
	Emily Grove
	Steve Hoffman

6. Write a query to list only those employees who have someone reporting to them. Also, show the number of reporters (including the President).

```

41 • SELECT
42     M.EMP_ID,
43     M.FIRST_NAME,
44     M.LAST_NAME,
45     COUNT(E.EMP_ID) AS NUMBER_OF_REPORTERS
46 FROM emp_record_table M
47 JOIN emp_record_table E
48     ON M.EMP_ID = E.MANAGER_ID
49 GROUP BY
50     M.EMP_ID, M.FIRST_NAME, M.LAST_NAME;

```

Result 13

EMP_ID	FIRST_NAME	LAST_NAME	NUMBER_OF_REPORTERS
E103	Emily	Grove	2
E428	Pete	Allen	3
E083	Patrick	Voltz	3
E001	Arthur	Black	5
E583	Janet	Hale	3
E612	Tracy	Marrie	2

7. Write a query to list down all the employees from the healthcare and finance departments using union. Take data from the employee record table.

```

60 WHERE DEPT = 'Finance';

```

Result 14

EMP_ID	FIRST_NAME	LAST_NAME	DEPT
E052	Dianna	Wilson	HEALTHCARE
E057	Dorothy	Wilson	HEALTHCARE
E083	Patrick	Voltz	HEALTHCARE
E505	Chad	Wilson	HEALTHCARE
E005	Eric	Hoffman	FINANCE
E103	Emily	Grove	FINANCE

8. Write a query to list down employee details such as EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPARTMENT, and EMP_RATING grouped by dept. Also include the respective employee rating along with the max emp rating for the department.

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EMP_RATING	MAX_DEPT_RATING
E001	Arthur	Black	PRESIDENT	ALL	5	5
E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	2	5
E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	5	5
E428	Pete	Allen	MANAGER	AUTOMOTIVE	4	5
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1	5
E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	3	4

9. Write a query to calculate the minimum and the maximum salary of the employees in each role. Take data from the employee record table.

```

73 • SELECT
74     ROLE,
75     MIN(SALARY) AS MIN_SALARY,
76     MAX(SALARY) AS MAX_SALARY
77 FROM emp_record_table
78 GROUP BY ROLE;

```

ROLE	MIN_SALARY	MAX_SALARY
PRESIDENT	16500	16500
LEAD DATA SCIENTIST	8500	9000
SENIOR DATA SCIENTIST	5500	7700
MANAGER	8500	11000
ASSOCIATE DATA SCIENTIST	4000	5000
LEAD DATA SCIENTIST	2800	3000

10. Write a query to assign ranks to each employee based on their experience. Take data from the employee record table.

EMP_ID	FIRST_NAME	LAST_NAME	EXP	EMP_RANK
E001	Arthur	Black	20	1
E083	Patrick	Voltz	15	2
E103	Emily	Grove	14	3
E428	Pete	Allen	14	3
E583	Janet	Hale	14	3
E612	Tracy	Morris	13	6

11. Write a query to create a view that displays employees in various countries whose salary is more than six thousand. Take data from the employee record table.

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	EMP_ID	FIRST_NAME	LAST_NAME	COUNTRY	SALARY
▶	E001	Arthur	Black	USA	16500
	E005	Eric	Hoffman	USA	8500
	E010	William	Butler	FRANCE	9000
	E057	Dorothy	Wilson	USA	7700
	E083	Patrick	Voltz	USA	9500
	E103	Emily	Grove	CANADA	10500

high_salary_employees 18 x

12. Write a nested query to find employees with experience of more than ten years. Take data from the employee record table.

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	EMP_ID	FIRST_NAME	LAST_NAME	EXP
▶	E001	Arthur	Black	20
	E005	Eric	Hoffman	11
	E010	William	Butler	12
	E083	Patrick	Voltz	15
	E103	Emily	Grove	14
	E428	Dea	Allen	14

emp_record_table 19 x

13. Write a query to create a stored procedure to retrieve the details of the employees whose experience is more than three years. Take data from the employee record table.

125 • SELECT * FROM emp_record_table WHERE EXP > 3;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	P_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER
▶	1	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL
	5	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103
	0	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428
	2	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083
	7	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083

emp_record_table 20 x

Read Only

14. Write a query using stored functions in the project table to check whether the job profile assigned to each employee in the data science team matches the organization's set standard.

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
CheckJobProfile('Data Scientist')			
Match			

Result 21 x

15. Create an index to improve the cost and performance of the query to find the employee whose FIRST_NAME is 'Eric' in the employee table after checking the execution plan.

199 emp_id,

< >

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered
▶		SIMPLE	emp_record_table	NULL	ref	idx_employee_first_name,idx_employee_fname...	idx_employee_first_name	403	const	1	100.00

< >

Result 23 

Output

Read Only 

16. Write a query to calculate the bonus for all the employees, based on their ratings and salaries (Use the formula: 5% of salary * employee rating).

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	
emp_id	first_name	salary	emp_rating	bonus
E001	Arthur	16500	5	4125.00
E005	Eric	8500	3	1275.00
E010	William	9000	2	900.00
E052	Dianna	5500	5	1375.00
E057	Dorothy	7700	1	385.00
E088	Patrick	9500	5	2375.00

Result 24 x

Output

17. Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table.

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	continent	country	avg_salary
▶	ASIA	CHINA	6500.0000
	ASIA	INDIA	6166.6667
	EUROPE	FRANCE	9000.0000
	EUROPE	GERMANY	7600.0000
	NORTH AMERICA	CANADA	7000.0000
	NORTH AMERICA	USA	9440.0000

Result 25 x

Output